

EdgeWake

Tiered Cascade Guardian

Ultra-Low-Power Forest Surveillance

"Our system only speaks when spoken to by a disaster."



The Problem

Why Forest Sensors Fail

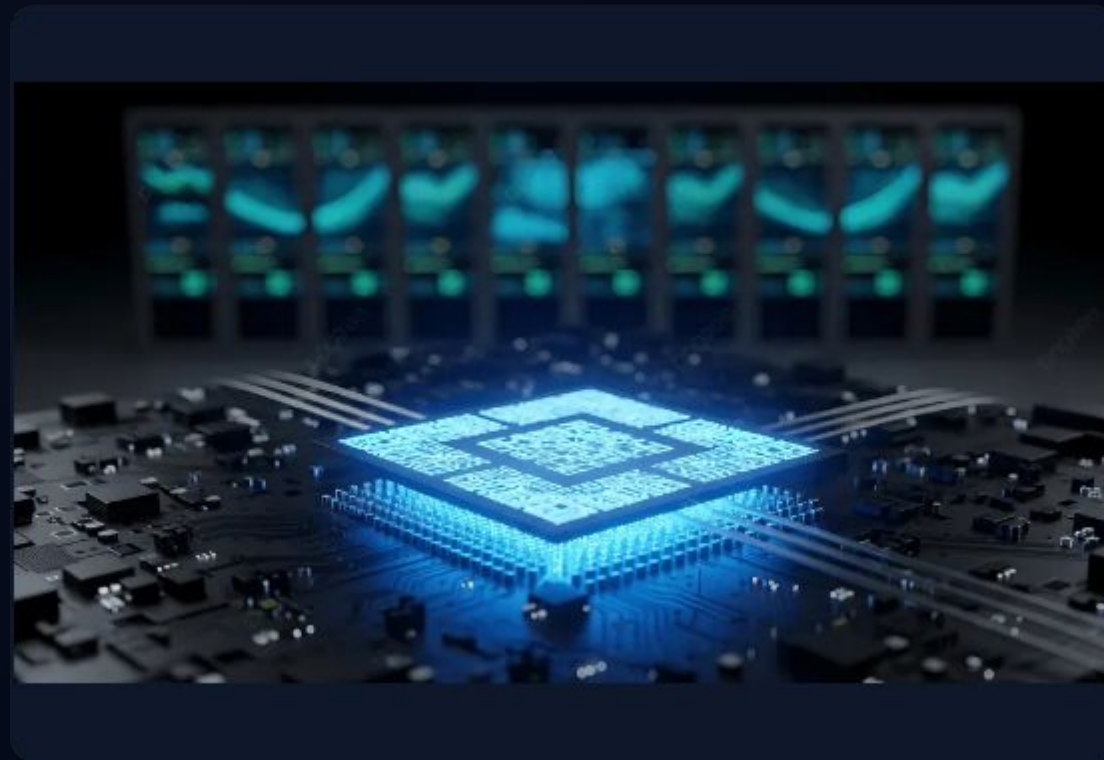
- Forest fires destroy **10+ million hectares** globally every year.
 - Illegal logging and poaching go undetected in remote, unmonitored areas.
 - Existing IoT sensors rely on fixed-schedule cloud communication.
 - Constant Wi-Fi and sensor polling drains batteries in days, not months.
 - Replacing batteries in dense forests is highly impractical and expensive.
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- **Result: Sensors die long before they detect a real threat.**



| EdgeWake — The Solution

An ultra-low-power, multi-modal edge AI node that sleeps 99% of the time.

- Instead of constantly polling the cloud, EdgeWake uses low-power analog tripwires (flame sensor, vibration sensor).
- These analog sensors **WAKE UP** high-power systems (microphone, camera, Wi-Fi) ONLY when a real threat is detected.
- The device spends 99% of its life in deep sleep ($\sim 10 \mu\text{A}$) and only uses full power ($\sim 160 \text{ mA}$) for 5 seconds during an actual alert.
- **Battery life:** Extended to months or years instead of mere days.



| What Makes EdgeWake Different?

Feature	Traditional Sensor	EdgeWake System
Power Mode	Always ON	Deep Sleep (99%)
Cloud Comms	Every 30 seconds	Only on confirmed threat
Battery Life	2-7 days	6-18 months
False Alarms	Sends everything to cloud	3-stage Edge verification
Intelligence	Cloud-only processing	Edge AI on device
Sensors	Single mode	Multi-modal (IR + Audio + Vision)

The Tiered Cascade Architecture

5 tiers from zero-power to full-alert in under 5 seconds.



Tier 1 — Deep Sleep (Near-Zero Power)

Powered Down

The main ESP32 CPU, Wi-Fi radio, and camera module are physically powered OFF.

Only the ULP (Ultra Low Power) co-processor remains minimally active to monitor the environment.

Current Draw: ~10 μ A

Enough to last months on a single 18650 battery.

Analog Tripwires Active

Two critical analog sensors remain powered, waiting for anomalies:

→ **Flame Sensor:** (IR receiver) on GPIO 13

→ **Vibration Sensor:** (SW-420) on GPIO 2

The system uses ext1 hardware wake-up — the CPU does NOT poll. It is woken solely by a voltage spike on these sensor pins.

Tier 2 — Hardware Interrupt (The Tripwire)



The Trigger Event

An environmental anomaly occurs (e.g., flash of fire, illegal logging chainsaw, or landslide).

The flame sensor detects infrared radiation OR the vibration sensor detects abnormal ground vibration.

The sensor sends a HIGH voltage signal directly to an RTC GPIO pin.



Instant Wake-up

This triggers an ext1 hardware wake-up. The ESP32 powers on instantly.

Firmware identifies WHICH sensor fired by reading the ext1 bitmask:

- **GPIO 13 HIGH:** "Fire" alert
- **GPIO 2 HIGH:** "Abnormal Vibration"

Debounce logic prevents repeated false triggers.

Tier 3 — Edge Audio Verification

Before activating the power-hungry camera, the system verifies the threat using sound.

The INMP441 I2S microphone records 2 seconds of audio at 16 kHz.

Two Analysis Modes:

- **Simple:** RMS energy threshold (fast, effective).
- **TinyML:** Edge Impulse model classifies sounds as "fire", "chainsaw", or "background".

Threat Confirmed: Proceed to camera.

False Alarm: Back to sleep instantly.



| Tier 4 — Visual Evidence Capture



Irrefutable Proof

This tier is **only reached** if audio verification confirms a real threat exists, saving massive amounts of power over the device's lifetime.

- The OV2640 camera module powers on.
- Captures a single high-resolution JPEG (800×600).
- Flash LED fires briefly for illumination in low-light, dense forest conditions.
- Uses PSRAM for high-resolution frame buffers.

The photo serves as court-admissible evidence for illegal logging prosecution or fire insurance claims.

Tier 5 — Cloud Handoff & Instant Sleep

Burst Transmission

The ESP32 briefly connects to Wi-Fi (typically taking < 3 seconds).

Sends a multipart HTTP POST to an **n8n webhook** containing critical data:

- The JPEG evidence photo
- Alert type ("Fire" or "Vibration")
- Device ID and GPS/location
- Audio RMS energy value & Timestamp

Return to Zero

After the server responds with 200 OK, confirming receipt, Wi-Fi is immediately severed.

The device forces itself back into Tier 1 Deep Sleep within milliseconds.

Total Awake Time: ~5 seconds

The threat is reported, and the battery is protected.

| Cloud-Side Automation with n8n

All heavy processing is offloaded to the cloud to save ESP32 power.



1. Webhook

Receives the multipart POST alert from the EdgeWake ESP32 node.



2. Smart Router

Routes payload dynamically by alert type (Fire / Vibration / Intruder).



3. Secure Storage

Stores the high-res evidence photo in Drive with a timestamped filename.



4. Instant Alert

Pushes photo + emergency message directly to the user's phone.

Hardware Bill of Materials

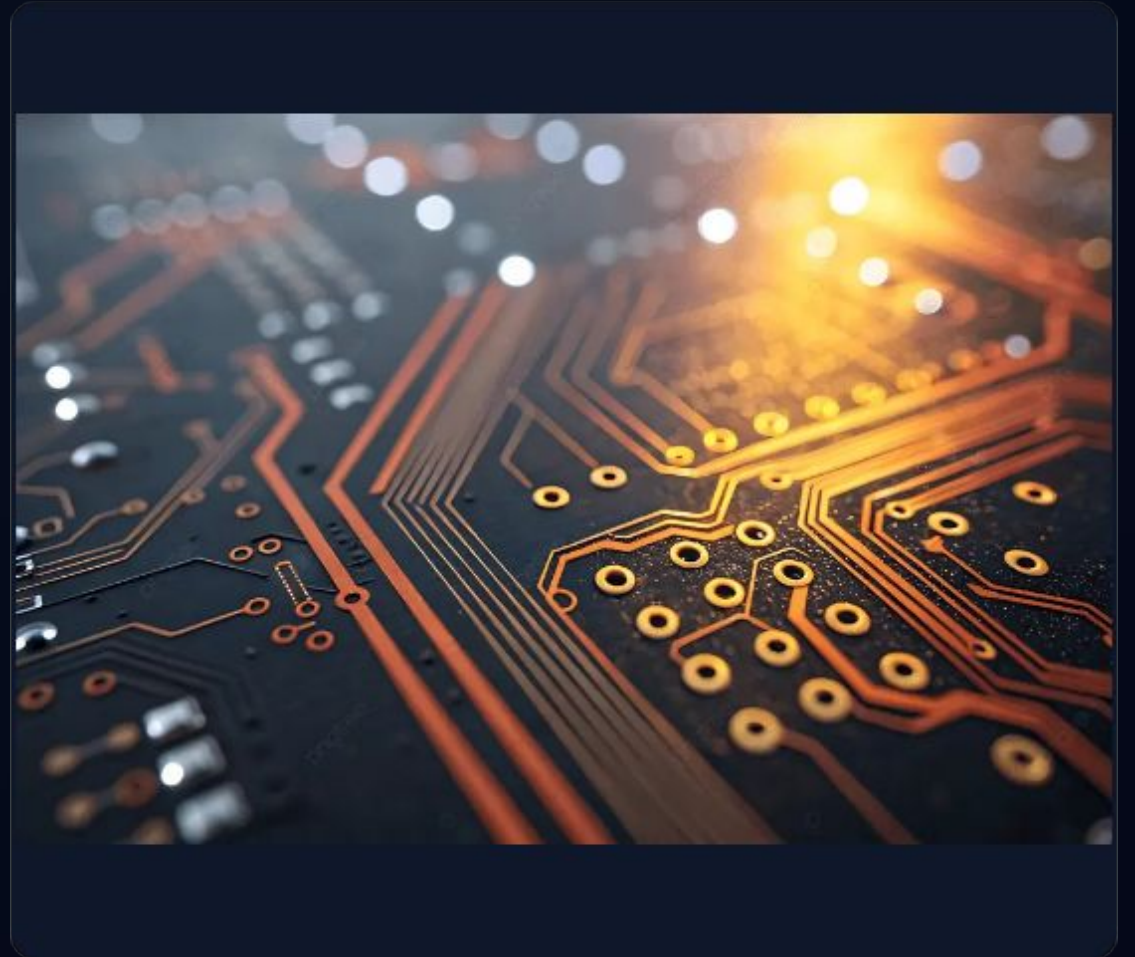
Component	Model	Purpose	Cost (approx.)
Microcontroller	ESP32-CAM (AI-Thinker)	Main brain + OV2640 camera	₹500
Programmer	FTDI USB-to-TTL (CP2102)	Upload firmware	₹150
Microphone	INMP441 I2S MEMS Mic	Audio verification (Tier 3)	₹200
Flame Sensor	Generic IR Flame Module	Fire detection tripwire (Tier 2)	₹50
Vibration Sensor	SW-420	Logging/movement tripwire	₹40
Power	18650 Li-ion + TP4056	Long-life battery supply	₹200

Wiring & Pin Assignments

Core Connections:

- Flame Sensor DO → **GPIO 13** (RTC ext1)
- Vibration Sensor DO → **GPIO 2** (RTC ext1)
- Camera: Built-in OV2640 (Internal)
- Power: 5V via 18650+TP4056+MT3608

GPIO	Function	Module
2	ext1 Wake	SW-420
4	Flash LED	Built-in
12	I2S Data In	INMP441
13	ext1 Wake	Flame Sensor
14	I2S Word Select	INMP441
15	I2S Clock	INMP441



Software & Technology Stack



Edge Layer (ESP32)

Arduino C++ (ESP-IDF)

- esp_sleep.h — deep sleep
- esp_camera.h — OV2640
- driver/i2s.h — INMP441 mic capture
- HTTPClient.h — multipart POST



AI Layer (TinyML)

Edge Impulse

- TensorFlow Lite for Microcontrollers
- MFCC feature extraction
- Neural classifier (Fire/Chainsaw)
- On-device inference optimization

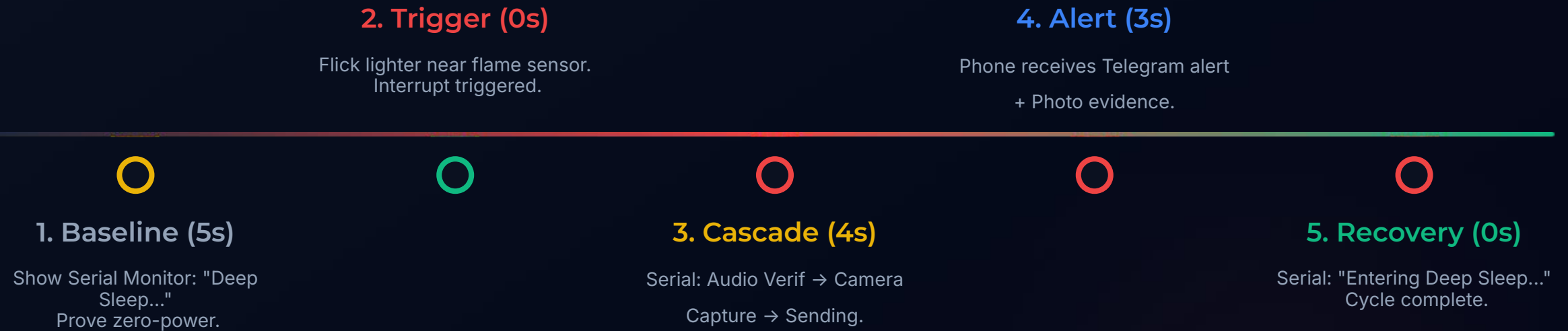


Cloud / App Layer

n8n & Web Dashboard

- n8n (open-source automation)
- HTML5/CSS3/JS Web Dashboard
- Real-time cascade simulation
- Telegram API integration

| Live Demo — 12-Second Proof



| Real-World Use Cases



Forest Fire Early Warning Detect nascent fires within seconds, preventing widespread destruction.



Illegal Logging Detection Vibration & audio models isolate and detect chainsaw and felling activity.



Anti-Poaching Surveillance Sound classification tailored for gunshot detection in reserves.



Landslide Monitoring Vibration tripwires detect subtle ground movements pre-collapse.



Industrial Safety Monitor unmanned facilities for fire or structural failures remotely.



Agricultural Monitoring Detect pest swarms via audio, and fire risk during dry seasons.

| Future Roadmap



LoRa / Satellite Mesh Eliminate Wi-Fi dependency for true deep-forest deployment.



Solar + Supercapacitor Transition to infinite battery life utilizing energy harvesting.



GPS Integration Auto-report exact threat coordinates directly to responders.



Multi-Node Dashboard Monitor hundreds of fleet nodes from a single master web panel.



On-Device Retraining Adapt AI models to new threat signatures without re-flashing.



IP67 Enclosure Ruggedized, weatherproof 3D-printed housing for severe climates.

EdgeWake

"The smartest sensor is the one that knows when NOT to work."

6–18 Mos

Battery Life

< 5s

Alert
Latency

< ₹1,200

Cost per Node

90%+

False Alarm Red.

Thank You

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